

Problem Set # 15

S12.6.1

$$\psi_E(r, \theta, \phi) = A e^{-r/a_0}$$

i. $l = 0$

ii.

$$\left(-\frac{\hbar^2}{2\mu} \left[\frac{1}{r^2} \frac{d}{dr} r^2 \frac{d}{dr} - \frac{l(l+1)}{r^2} \right] + V(r) \right) e^{-r/a_0} = E e^{-r/a_0}$$

← vanishes

$$V(r) \rightarrow 0 \quad \text{as} \quad r \rightarrow \infty$$

$$\left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) e^{-r/a_0} \rightarrow \frac{1}{a_0^2} e^{-r/a_0}$$

$$-\frac{\hbar^2}{2\mu} \frac{1}{a_0^2} = E$$

iii.

$$\left(-\frac{\hbar^2}{2\mu} \frac{2}{r} \frac{d}{dr} + V(r) \right) e^{-r/a_0} = 0$$

$$V(r) = -\frac{\hbar^2}{\mu a_0} \frac{1}{r} = -\frac{e^2}{r} \quad \left(\text{for } a_0 = \frac{\hbar^2}{\mu e^2} \right)$$

512.6.4

$$\begin{aligned}
 \text{a.} \quad & \int_0^{\infty} r^2 dr \int_0^{\pi} \sin \theta d\theta \int_0^{2\pi} d\varphi f(r, \theta, \varphi) \delta^3(\vec{r} - \vec{r}') \\
 &= \int_0^{\infty} r^2 dr \int_0^{\pi} \sin \theta d\theta \int_0^{2\pi} d\varphi f(r, \theta, \varphi) \underbrace{\delta(r-r') \delta(\theta-\theta') \delta(\varphi-\varphi')}_{r^2 \sin \theta} \\
 &= f(r', \theta', \varphi')
 \end{aligned}$$

$$\begin{aligned}
 \text{b.} \quad & \nabla^2 \left(\frac{1}{r} \right) = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} \frac{1}{r} \right) \\
 &= \frac{1}{r^2} \frac{d}{dr} \left(r^2 \left(-\frac{1}{r^2} \right) \right) = 0 \quad \text{for } r \neq 0
 \end{aligned}$$

$$\begin{aligned}
 \int_{r < R} d^3x \nabla^2 \left(\frac{1}{r} \right) &= \int_{r < R} d^3x \vec{\nabla} \cdot \left(\vec{\nabla} \frac{1}{r} \right) \\
 &= \int_{r=R} d^2x \vec{n} \cdot \vec{\nabla} \frac{1}{r} = -4\pi R^2 \frac{1}{R^2} = -4\pi
 \end{aligned}$$

$$\therefore \nabla^2 \left(\frac{1}{r} \right) = -4\pi \delta^3(\vec{r})$$

12.6.5

$$D_l = -\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2}$$

Consider two solutions U_1 and U_2
with the same l and E eigenvalues:

$$D_l U_1 = E U_1 \quad \text{and}$$

$$D_l U_2 = E U_2$$

Then:

$$+ \frac{\hbar^2}{2\mu} \left[U_2 \frac{d^2 U_1}{dr^2} - U_1 \frac{d^2 U_2}{dr^2} \right]$$

$$= \left[V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] (U_1 U_2 - U_2 U_1) = 0.$$

Hence

$$\frac{d}{dr} \left[U_2 \frac{dU_1}{dr} - U_1 \frac{dU_2}{dr} \right] = 0$$

$$\therefore U_2 \frac{dU_1}{dr} - U_1 \frac{dU_2}{dr} = c, \text{ a constant.}$$

$$U_{1,2} \rightarrow 0 \quad \text{as } r \rightarrow 0$$

$$\text{Let } U_j \rightarrow c_j r^{\alpha_j} \quad \text{as } r \rightarrow 0$$

$$j = 1, 2 \quad \alpha_j > 0$$

$$C = \lim_{r \rightarrow 0} U_2 \frac{dU_1}{dr} - U_1 \frac{dU_2}{dr} = c_1 c_2 r^{\alpha_1 + \alpha_2 - 1} (\alpha_1 - \alpha_2)$$

$$\text{For } l \geq 1, \quad \alpha_1 = \alpha_2 = l + 1$$

(cfr. Shankar, p 352), Hence $C = 0$.

For $l = 0$, if V is less singular than r^{-2} as $r \rightarrow 0$,

then $U \rightarrow r^\alpha$ with $\alpha = 0$ or 1 ,

by the same argument as on page 352

of Shankar's book, $\alpha = 0$ is clearly

not allowed (cfr page 351 of Shankar).

Thus $\alpha_1 = \alpha_2 = 1$. Hence $C = 0$.

In all cases that I am aware of, the boundary conditions at $r=0$ require $\xi=0$. Hence

$$U_2 \frac{dU_1}{dr} - U_1 \frac{dU_2}{dr} = 0$$

$$\therefore \frac{d}{dr} \left(\frac{U_1}{U_2} \right) = 0$$

$$\therefore U_1 = \text{constant} \times U_2$$

Hence U_1 and U_2 describe the same state.

Hence for each set of values for E , l and m , there is at most one state (whether $E > 0$ or $E < 0$).

512.6.8

$$\psi_{Elm}(r, \theta, \phi) = R_{El}(r) Y_e^m(\theta, \phi)$$

$$\frac{-\hbar^2}{2\mu} \left[\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{l(l+1)}{r^2} \right] R_{El}(r) = E R_{El}(r)$$

$$R_{El}(r) \sim r^l \quad \text{as } r \rightarrow 0$$

$$[\text{For } l=0, R_{El}(r) \sim \frac{1}{r} \text{ is}$$

not allowed because

$$\nabla^2 \frac{1}{r} = -4\pi \delta^3(\vec{x}) \neq 0.$$

For $l \neq 0$, see discussion on p. 352]

$$R_{El}(r) = N j_l(kr) \quad E = \frac{\hbar^2 k^2}{2\mu}$$

$$= N (-p)^l \left(\frac{1}{p} \frac{d}{dp} \right)^l \frac{\sin p}{p} \quad | \quad p = kr$$

The energy eigenvalues $E_{lm} = \frac{\hbar^2 k_{lm}^2}{2\mu}$ are determined by the boundary condition

$$\psi_{Elm}(r_0, \theta, \phi) = 0$$

$\therefore k_{lm}$ is such that $j_l(k_{lm} r_0) = 0$

For $l = 0$

$$j_0(k_n r_0) = \frac{\sin(k_n r_0)}{k_n r_0} = 0$$

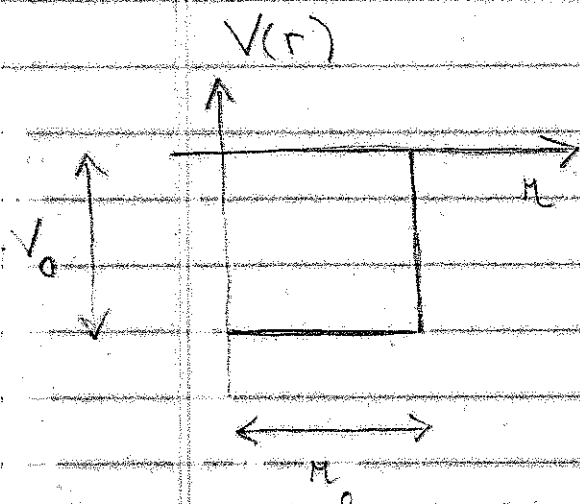
$$k_n r_0 = m\pi \quad m = 1, 2, 3, \dots$$

$$E_m = \frac{\hbar^2 \pi^2}{2\mu} \frac{m^2}{r_0^2}$$

$$\psi_{m00} = N_m \frac{1}{r} \sin\left(\frac{m\pi}{r_0} r\right)$$

5.12.6.9

$$\psi_{Elm}(r, \theta, \varphi) = R_{El}(r) Y_l^m(\theta, \varphi)$$



$$\left[-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{l(l+1)}{r^2} \right) \right.$$

$$\left. + V(r) \right] R_{El}(r) = E R_{El}(r)$$

($V_0 < 0$)

$$R_{El}(r) = \frac{1}{r} U_{El}(r), \quad E < 0$$

$$\left[\frac{d^2}{dr^2} - \frac{l(l+1)}{r^2} + k^2 \right] U_{El}(r) = 0 \quad \text{for } r \leq r_0$$

$$\left[\frac{d^2}{dr^2} - \frac{l(l+1)}{r^2} - \kappa^2 \right] U_{El}(r) = 0 \quad \text{for } r > r_0$$

with $E - V_0 = \frac{\hbar^2 k^2}{2\mu}$ and $-E = \frac{\hbar^2 \kappa^2}{2\mu}$

$U_{El}(r)$ and $\frac{dU_{El}}{dr}$ must be continuous at $r = r_0$.

For $l=0$

$$U(r) = A \sin kr \quad \text{for } r < r_0$$

$$= B e^{-\kappa r} \quad \text{for } r > r_0$$

($U(r) \sim \cos kr$ for $r < r_0$ does not solve the Schrödinger equation because $\nabla^2 \frac{1}{r} = -4\pi \delta^3(\vec{r}) \neq 0$)

$$\begin{cases} A \sin kr_0 = B e^{-\kappa r_0} \\ A k \cos kr_0 = -B \kappa e^{-\kappa r_0} \end{cases}$$

$$\tan kr_0 = -k/\kappa$$

determines the discrete spectrum of bound state energies

$$\kappa^2 = 2\mu \left(\frac{-E}{\hbar^2} \right)$$

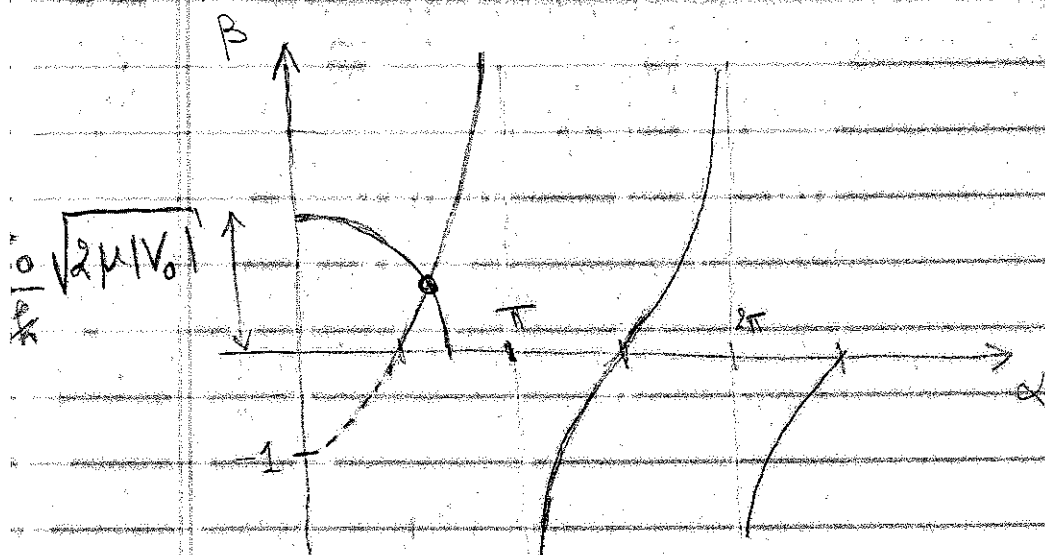
$$k^2 = 2\mu \left(\frac{E - V_0}{\hbar^2} \right)$$

$$\kappa^2 + k^2 = \frac{2\mu}{\hbar^2} (-V_0)$$

Let $kr_0 \equiv \beta$ and $k r_0 \equiv \alpha$

$$\alpha^2 + \beta^2 = \frac{2\mu}{\hbar^2} r_0^2 |V_0|$$

$$\beta = -\frac{\alpha}{\tan \alpha}$$



There are no bound states for

$$\frac{r_0}{\hbar} \sqrt{2\mu |V_0|} < \frac{\pi}{2}$$

or $|V_0| < \frac{\pi^2 \hbar^2}{8\mu r_0^2}$

12.6.10

$$e^{ikr \cos \theta} = \sum_{l=0}^{\infty} c_l P_l(\cos \theta) j_l(kr)$$

$$\int_{-1}^{+1} d(\cos \theta) P_l(\cos \theta) e^{ikr \cos \theta} d\cos \theta$$

$$= c_l \frac{2}{2l+1} j_l(kr)$$

$$= \int_{-1}^{+1} dx \frac{1}{2^l l!} \frac{d^l (x^2-1)^l}{dx^l} e^{ikrx}$$

$$= \frac{1}{2^l l!} (-ikr)^l \int_{-1}^{+1} dx (x^2-1)^l e^{ikrx}$$

by integrating by parts l times

$$= \frac{2}{2^l l!} (ikr)^l \int_0^1 dx (1-x^2)^l e^{ikrx}$$

Take the limit $kr \rightarrow 0$ on
both sides

$$j_l(kr) \rightarrow \frac{(kr)^l}{(2l+1)!!} \quad \text{as } kr \rightarrow 0$$

$$e^{ikr} \rightarrow 1 \quad \text{as } kr \rightarrow 0$$

$$C_l = \frac{2}{2l+1} \frac{1}{(2l+1)!!}$$

$$= \frac{2}{2^l l!} i^l \int_0^1 dx (1-x^2)^l$$

$$= \frac{2}{2^l l!} i^l \frac{(2l)!!}{(2l+1)!!}$$

$$\text{Hence } C_l = (2l+1) i^l \frac{(2l)!!}{2^l l!}$$

$$= (2l+1) i^l \quad \text{since}$$

$$(2l)!! = (2l)(2l-2) \dots 4 \cdot 2$$

$$= l(l-1) \dots 2 \cdot 1 \cdot 2^l$$

$$= l! \cdot 2^l$$